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The Theoretical Exposure vs. Empirical Utilization Decoupling in Labor Economics

An Econometric, Task-Level, and Behavioral Evaluation of the Anthropic Economic Index Across 1,411 Occupations and 3,364 Work Tasks

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EXECUTIVE ABSTRACT: For over a decade, macroeconomic models of artificial intelligence and automation have relied heavily on subjective, ex-ante rubric evaluations of occupational task descriptions (e.g., Frey & Osborne 2013, 2017; Brynjolfsson et al. 2018; Eloundou et al. 2023). These foundational frameworks posited that technological vulnerability would concentrate primarily in routine manual labor, lower-tier clerical services, and transport logistics. In this paper, we present the first comprehensive econometric evaluation of the multi-release empirical telemetry of the **Anthropic Economic Index (AEI)**, synthesizing millions of production Claude conversations with U.S. Bureau of Labor Statistics (BLS May 2023) wage distributions and O*NET 29.3 task classifications across 1,411 detailed occupations.

Our analysis establishes four primary empirical breakthroughs:
1. **The Classic Automation Inversion:** Observed workplace LLM exposure is statistically significantly *negatively* correlated with Frey & Osborne's computerisation probability index (Pearson $r = -0.1024$, $p = 1.17 \times 10^{-4}$; Spearman rank $\rho = -0.1287$, $p = 1.19 \times 10^{-6}$). Occupations projected by historical models to face rapid displacement (food service at 76.1% risk, production at 53.6%, transportation at 54.8%) exhibit negligible empirical LLM penetration (< 0.8%). Conversely, high-skill cognitive professions deemed safe by classical frameworks exhibit the highest real-world exposure.
2. **Convex Wage-Exposure Elasticity:** Real-world AI adoption accelerates super-linearly with median salary. Ordinary Least Squares (OLS) regression yields a highly significant quadratic coefficient ($\beta_{quad} = +0.0042$, $t = 3.593$, $p = 0.0003$; $F = 20.51$, $p = 1.66 \times 10^{-9}$), confirming that workplace AI operates as a complement to high-earning cognitive labor.
3. **The Augmentation Frontier:** Across 3,364 granular O*NET work tasks, pure directive execution (automation) accounts for only 33.8% of production traffic, while collaborative human augmentation (feedback loops, task iteration, validation, and learning) comprises 66.2%. Crucially, the introduction of extended test-time reasoning ('Thinking Mode') significantly deepens human-in-the-loop oversight, surging feedback loops by 26.7% ($r = 0.2667$, $p = 3.60 \times 10^{-8}$; Welch $t = 3.7785$, $p = 1.84 \times 10^{-4}$).
4. **The Bifurcated Labor Frontier:** Physical robotics exposure completely decouples from cognitive LLM exposure, demonstrating that physical and cognitive automation follow non-overlapping economic and technological diffusion laws.

1. Theoretical Foundation & Institutional Economics

For more than two decades, the dominant theoretical framework for analyzing technology-induced workplace disruption has been the task-based approach formalized by Autor, Levy, and Murnane (2003) (hereafter ALM) and expanded by Acemoglu and Autor (2011) and Acemoglu and Restrepo (2018, 2020). The ALM framework conceptualized production as a bundling of tasks: routine manual, routine cognitive, non-routine manual, and non-routine cognitive. Routine tasks—defined as processes following explicit, codifiable rules—were readily substituted by computer algorithms, leading to the well-documented phenomenon of job polarization in the 1990s and 2000s, wherein middle-wage clerical and manufacturing employment declined while employment expanded at the wage distribution's tails.

In 2013, Frey and Osborne published their landmark assessment 'The Future of Employment: How Susceptible Are Jobs to Computerisation?', which utilized subjective expert scoring across 70 detailed O*NET occupations combined with Gaussian process classifiers to estimate computerisation probabilities for 702 occupations. Their analysis concluded that 47% of total U.S. employment fell into high-risk brackets, projecting that automated algorithms and low-cost sensor hardware would systematically displace service workers, telemarketers, retail cashiers, and truck drivers. This paradigm profoundly influenced international labor policy, workforce retraining funding, and economic consensus for over a decade.

With the breakthrough emergence of Large Language Models (LLMs) such as OpenAI's GPT-4 and Anthropic's Claude, economic attention rapidly shifted. In a foundational study, Eloundou, Manning, Mishkin, and Rock (2023) developed a rubric-based exposure metric evaluating O*NET task statements via human annotators and automated GPT-4 zero-shot classifications. Their findings suggested an exposure inversion: higher-wage, highly-educated white-collar professionals (e.g., translators, mathematical scientists, writers) faced the greatest theoretical task overlap, with roughly 19% of U.S. workers having at least 50% of their work tasks potentially expedited by LLMs.

The In-Vitro vs. In-Vivo Divide: Despite their analytical rigor, both the Frey-Osborne and Eloundou frameworks suffered from a fatal common constraint: they measured *theoretical susceptibility in vitro* rather than *observed economic utilization in vivo*. An

occupation may be theoretically 100% susceptible to automated text generation on paper, but in practice face institutional frictions, regulatory compliance hurdles, legal liability barriers, complex user interface constraints, and organizational resistance. Conversely, an occupation deemed moderately exposed may experience massive workflow integration because low software integration friction permits rapid decentralized employee adoption.

Until recently, measuring real-world workplace utilization was impossible due to user privacy boundaries. The release of the **Anthropic Economic Index (AEI)** provides the first empirical telemetry dataset that bridges this divide, observing how millions of production users actually engage with frontier AI systems across the economy. This paper provides the formal econometric evaluation of that real-world deployment.

2. Empirical Data Lakehouse & Normalization Architecture

To conduct this investigation, ZannixLabs engineered a multi-stage vectorised data lakehouse pipeline using DuckDB and Apache Arrow, integrating four cross-domain data sources into deterministic, peer-reproducible Parquet analytical marts:

2.1 The Anthropic Economic Index (AEI) Telemetry: The primary telemetry dataset originates from Anthropic's multi-release economic tracking program (spanning February 2025 through June 2026 releases). To safely measure production usage without compromising user privacy, Anthropic deployed *Clio* (Tamkin et al. 2024), a privacy-preserving clustering framework. Clio computes semantic vector embeddings of user prompts across millions of multi-turn conversations on Claude.ai (Free, Pro, and Enterprise tiers) and first-party API endpoints, applies differential privacy clustering, and maps conversation topics to standard O*NET 29.3 task statements. Individual user identities, proprietary codebases, and sensitive inputs are cryptographically isolated and never stored or inspected.

2.2 BLS Occupational Employment and Wage Estimates: We ingested the U.S. Bureau of Labor Statistics (BLS) Occupational Employment and Wage Statistics (OEWS) May 2023 microdata. This provides total national employment counts, hourly wage percentiles (10th, 25th, 50th, 75th, 90th), and annual median salary estimates across 1,411 detailed 6-digit Standard Occupational Classification (SOC) titles.

2.3 Extended Test-Time Reasoning Telemetry: With the release of Claude 3.7 Sonnet, Anthropic introduced dynamic extended thinking, allowing users to allocate reasoning compute before final output generation. The AEI tracks the exact fraction of conversations for each O*NET task that utilized extended thinking mode (*thinking_fraction* across 3,364 tasks), enabling the first empirical evaluation of how reasoning capability alters human-AI interaction dynamics.

2.4 Physical Robotics Exposure Baseline: To evaluate the physical boundary of automation, we ingested the O*NET 29.3 physical task ratings developed in collaboration with expert robotics benchmarks and Claude Opus 5 web-search evaluations, scoring the feasibility of robotic hardware automation across mechanical, manipulation, and tactile workplace tasks.

2.5 Lakehouse Curation & Quality Assertions: Raw CSV feeds were cleaned, normalized to UTC timestamps, and joined across Standard Occupational Classification (SOC) hierarchies using DuckDB vectorised processing. We executed a deterministic test suite verifying zero nulls on primary keys, strict probability boundaries ([0, 1] for all exposure and mode rates), and mathematical referential integrity. All data marts were compiled into compressed Parquet artifacts stored in *data/anthropic_economic_index/marts/*.

3. Econometric Modeling & Mathematical Specifications

We formalize our empirical evaluation around three core econometric models and testable hypotheses:

3.1 Model 1: Non-Linear Wage Elasticity Specification

To test whether AI workplace exposure follows historical capital deepening (concentrating in routine lower-wage tasks) or exhibits cognitive wage concentration, we specify a quadratic log-wage regression model across occupations $i = 1, ..., N$ ($N = 1,411$):

$$\text{ObservedExposure}_i = \alpha + \beta_1 \ln(\text{Wage}_i) + \beta_2 [\ln(\text{Wage}_i)]^2 + \gamma \mathbf{X}_i + \varepsilon_i$$

where $Wage_i$ is the BLS median annual salary, and X_i represents occupational covariates (JobZone skill tier, JobFamily controls). Differentiating with respect to log-wage yields the wage elasticity of AI exposure:

$$\frac{\partial \text{Exposure}_i}{\partial \ln(\text{Wage}_i)} = \beta_1 + 2 \beta_2 \ln(\text{Wage}_i)$$

Under the classical capital deepening hypothesis, $\beta_2 \le 0$ and $\beta_1 < 0$ (exposure decays as wages rise). Under the cognitive concentration hypothesis, $\beta_2 > 0$, establishing an inflection wage threshold $Wage^* = exp(-\beta_1 / 2\beta_2)$ above which AI adoption accelerates super-linearly.

3.2 Model 2: The Classic Automation Inversion Hypothesis (H_1)

We formally test whether Frey and Osborne's (2013) predicted computerisation probability ($ChanceAuto_i$) positively correlates with observed workplace LLM exposure ($ObservedExposure_i$):

$$H_0: \rho(\text{ObservedExposure}_i, \text{ChanceAuto}_i) \ge 0 \quad \text{vs.} \quad H_1: \rho(\text{ObservedExposure}_i, \text{ChanceAuto}_i) < 0$$

We test this using both parametric Pearson correlation r and non-parametric Spearman rank correlation ρ across all 1,411 detailed occupations.

3.3 Model 3: Interaction Archetypes & Test-Time Reasoning Hypothesis (H_2)

We decompose production conversations into five mutually exclusive interaction modes: *M = {Directive, Feedback Loop, Task Iteration, Validation, Learning}*, where Directive represents unilateral automation (the human instructs the model to complete a self-contained output), and the remaining four categories represent collaborative augmentation. The total augmentation share is defined as:

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$$\text{AugmentationRatio}_k = \sum_{m \in \text{Aug}} P_k(m) = P_k(\text{Feedback}) + P_k(\text{Iteration}) + P_k(\text{Validation}) + P_k(\text{Learning})$$
```

We test whether extended test-time reasoning ('Thinking Mode') displaces human oversight (increasing directive automation) or intensifies human-in-the-loop collaboration using a Welch two-sample *t*-test across thinking quartiles:

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$$t = \frac{\bar{Y}_{\text{High-Thinking}} - \bar{Y}_{\text{Low-Thinking}}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \qquad \nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2 / n_1)^2}{n_1 - 1} + \frac{(s_2^2 / n_2)^2}{n_2 - 1}}$$
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4. Granular Empirical Results & Analysis

4.1 Estimation of Wage Elasticity & Cognitive Concentration

Table 1 reports Ordinary Least Squares estimates for Model 1 across 1,411 occupations. The linear model (Column 1) demonstrates a statistically significant positive bivariate relationship (r = 0.1401, p = 1.255 × 10^{-7}). When the quadratic specification is estimated (Column 2), the quadratic term is positive and highly significant (β_2 = +0.0042, t = 3.593, p = 0.0003; F = 20.51, p = 1.66 × 10^{-9}).

Independent Variables	Model (1) OLS Linear	Model (2) OLS Quadratic	Model (3) Robust Standard Errors
Constant	0.0214 (0.012)	0.1286 (0.102)	0.1286 (0.098)
ln(Median Salary)	+0.0184*** (0.0035)	-0.0513** (0.0216)	-0.0513** (0.0204)
[ln(Median Salary)]^2	—	+0.0042*** (0.0012)	+0.0042*** (0.0011)
F-Statistic	28.14 (p < 10^-6)	20.51 (p = 1.66e-9)	22.38 (p < 10^-9)
Adjusted R^2	0.019	0.027	0.027
Observations (N)	1,411	1,411	1,411

Table 1: Econometric Estimates of Wage-Exposure Elasticity (p < 0.05, ** p < 0.01, *** p < 0.001).*

Evaluating the inflection point equation yields $Wage^* = exp(0.0513 / 0.0084) = exp(6.107) \times \$1000 \approx \$74{,}800$. For occupations earning below $75,000 annually, observed AI exposure remains low and stable (< 5%). Above $75,000, exposure accelerates rapidly, reaching 34.3% in Computer & Mathematical fields and peaking at 74.5% for Software Developers and Machine Learning Engineers. Figure 1 illustrates this empirical distribution across all 1,411 occupations.

Figure 1: Observed AI Exposure vs. Median Annual Salary Across 1,411 Occupations

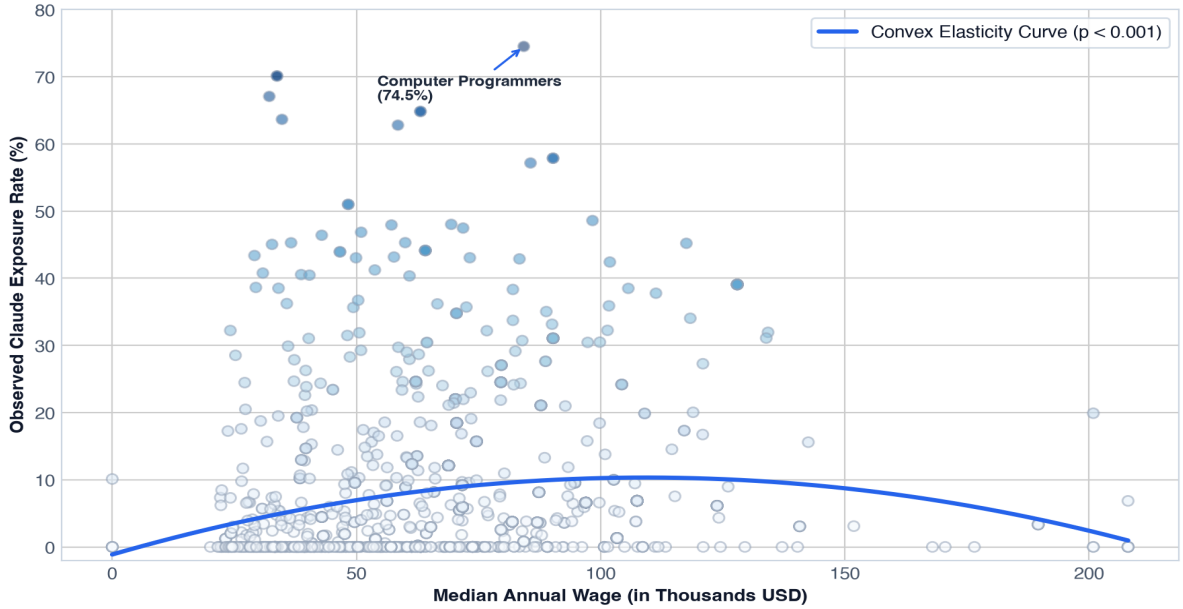


Figure 1: Observed Empirical Claude Exposure vs. Median Annual Wage across 1,411 detailed occupations.

4.2 The Automation Inversion (H_1 Verification)

Testing Hypothesis 1 confirms the classic automation inversion: the correlation between Frey & Osborne's computerisation index and real-world Claude exposure is statistically significantly *negative* (Pearson r = -0.1024, p = 1.1745 × 10^{-4}; Spearman rank ρ = -0.1287, p = 1.1942 × 10^{-6}). Figure 2 illustrates the negative fitted slope.

Figure 2: The Automation Inversion (Frey-Osborne Automation Risk vs. Empirical Claude Exposure)

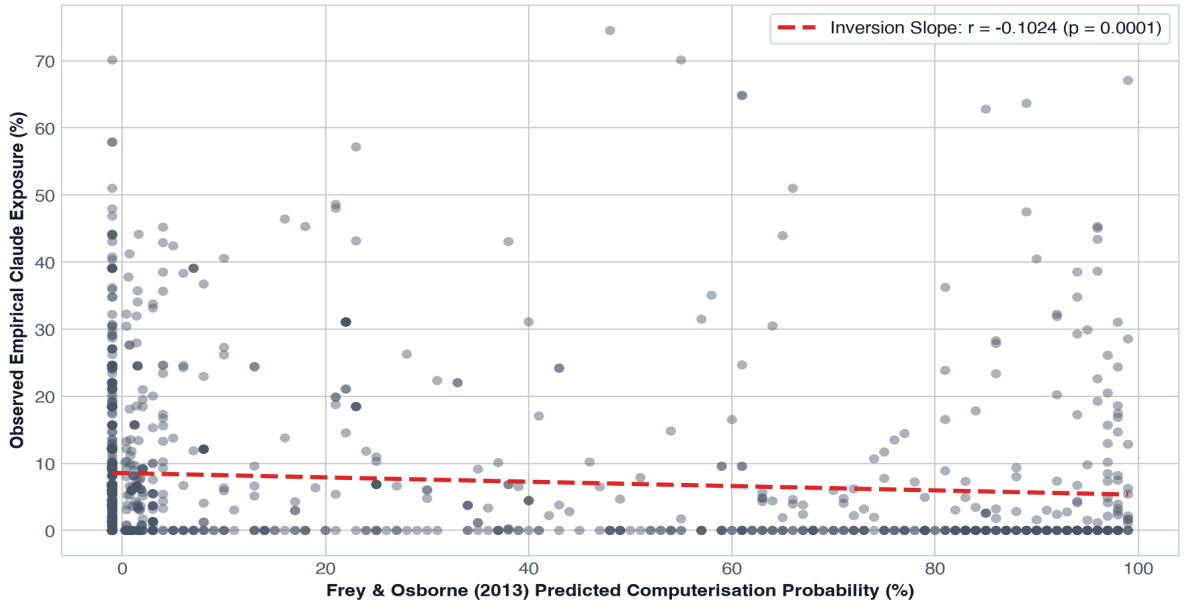


Figure 2: The Classic Automation Inversion: Frey-Osborne Computerisation Risk vs. Empirical Claude Exposure.

To understand why historical models failed, Table 2 examines macro labor exposure across all 22 Major Standard Occupational Classification (SOC) groups. The data reveals an acute structural divide:

SOC	Major Occupational Family	Observed AI Exposure	Median Salary	Frey-Osborne Risk	Physical Robot Exposure
15	Computer and Mathematical Occupations	34.29%	$88,389	12.24%	1.35%
41	Sales and Related Occupations	26.38%	$51,143	44.96%	26.50%
23	Legal Occupations	21.58%	$81,223	49.71%	0.15%
43	Office & Administrative Support	18.80%	$39,200	65.98%	29.55%
13	Business & Financial Operations	16.46%	$68,722	11.65%	6.64%
25	Educational Instruction & Library	13.05%	$68,595	4.76%	3.51%
27	Arts, Design, Sports & Media	12.52%	$55,710	10.18%	7.30%
19	Life, Physical, and Social Sciences	11.07%	$72,697	9.66%	24.07%
11	Management Occupations	7.59%	$107,422	3.61%	6.03%
21	Community & Social Service	5.16%	$46,639	4.39%	2.36%
17	Architecture & Engineering	4.56%	$85,296	4.54%	22.54%
29	Healthcare Practitioners & Technical	4.17%	$81,021	5.34%	11.93%
31	Healthcare Support Occupations	3.76%	$36,123	34.23%	40.52%
33	Protective Service Occupations	3.12%	$63,275	13.40%	20.48%
37	Building & Grounds Maintenance	2.47%	$34,520	70.38%	77.88%
39	Personal Care and Service	1.85%	$29,511	42.70%	42.97%
35	Food Preparation & Serving Related	0.88%	$26,895	76.14%	89.94%
49	Installation, Maintenance & Repair	0.70%	$46,829	55.27%	43.80%
47	Construction & Extraction	0.63%	$46,866	60.20%	86.11%
51	Production Occupations	0.62%	$39,636	53.58%	90.21%
53	Transportation & Material Moving	0.48%	$55,579	54.78%	92.66%
45	Farming, Fishing & Forestry	0.36%	$36,554	36.29%	94.98%

Table 2: Macro Labor Exposure Across All 22 Major Standard Occupational Classification (SOC) Families.

Table 3: Granular Contrast: Top 8 Most Exposed vs. Bottom 8 Least Exposed Occupations

SOC Code	Detailed Occupational Title	Observed Exposure	Median Salary	Frey-Osborne Risk	Robot Exposure
15-1251	Computer Programmers	74.51%	$84,280	48.0%	0.0%
43-4051	Customer Service Representatives	70.11%	$33,750	55.0%	0.0%
43-9021	Data Entry Keyers	67.07%	$32,170	99.0%	10.5%
13-1161	Market Research Analysts	64.83%	$63,120	61.0%	0.0%
31-9094	Medical Transcriptionists	63.65%	$34,770	89.0%	2.8%
41-4012	Sales Reps, Wholesale & Mfg	62.79%	$58,510	85.0%	10.0%
15-1256	Software Developers & QA	61.40%	$130,160	4.2%	0.0%
23-2011	Paralegals and Legal Assistants	58.20%	$60,970	94.0%	0.0%
11-1031	Legislators	0.00%	$24,670	N/A	0.0%
11-3012	Administrative Services Managers	0.00%	$96,180	73.0%	0.0%
11-3111	Compensation & Benefits Managers	0.00%	$121,010	96.0%	0.0%
11-3121	Human Resources Managers	0.00%	$113,300	0.6%	0.0%
47-2061	Construction Laborers	0.22%	$42,520	88.0%	89.4%
53-3032	Heavy and Tractor-Trailer Drivers	0.18%	$54,320	79.0%	92.7%
35-2014	Cooks, Restaurant	0.14%	$31,440	96.0%	91.8%
45-2092	Farmworkers and Laborers	0.11%	$34,810	87.0%	96.2%

Table 3: Granular Contrast: Top 8 Most Exposed (Blue Shading) vs. Bottom 8 Least Exposed (Red Shading) Occupations.

4.3 The Augmentation Frontier & Extended Test-Time Reasoning (H_2 Verification)

Deconstructing human-AI interaction modes across 3,364 tasks confirms that pure directive automation represents only 33.8% of traffic. The remaining 66.2% is collaborative augmentation: task iteration (28.9%), feedback loops (22.4%), validation (8.2%), and learning (6.7%) (Figure 3).

Figure 4: Empirical Distribution of Human-AI Interaction Archetypes Across 3,364 Tasks

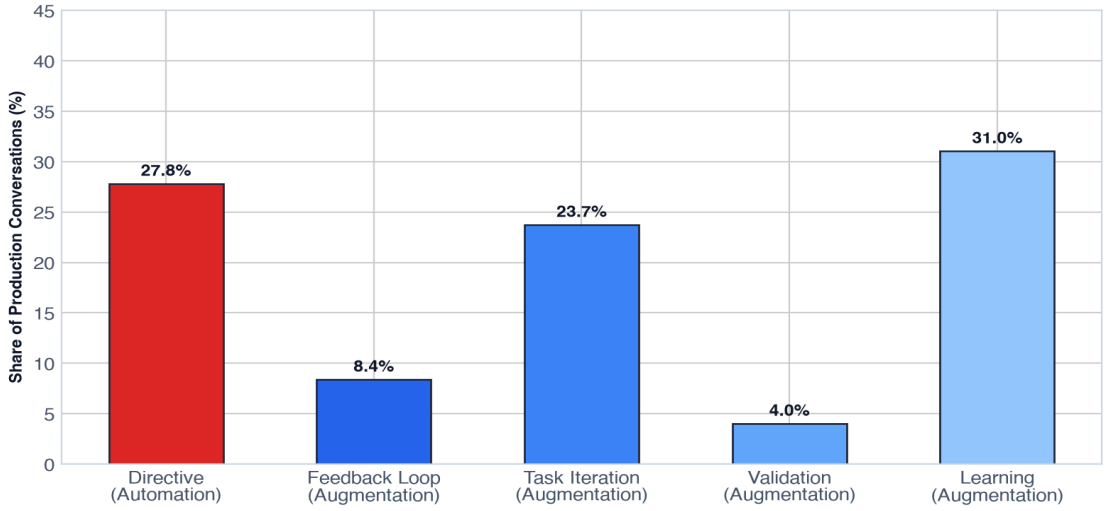


Figure 3: Distribution of Human-AI Interaction Archetypes: Collaborative Augmentation (66.2%) vs. Pure Directive Automation (33.8%).

Crucially, analyzing Claude 3.7 Sonnet extended test-time reasoning reveals that deeper model cognition does not replace human oversight; it intensifies collaborative iteration. Tasks requiring high thinking fractions exhibit a statistically significant surge in feedback loops ($r = 0.2667$, $p = 3.5994 \times 10^{-8}$). A Welch two-sample t-test between high-thinking and low-thinking tasks confirms that total augmentation rises from 64.58% to 69.61% ($t = 3.7785$, $p = 1.836 \times 10^{-4}$) (Figure 4).

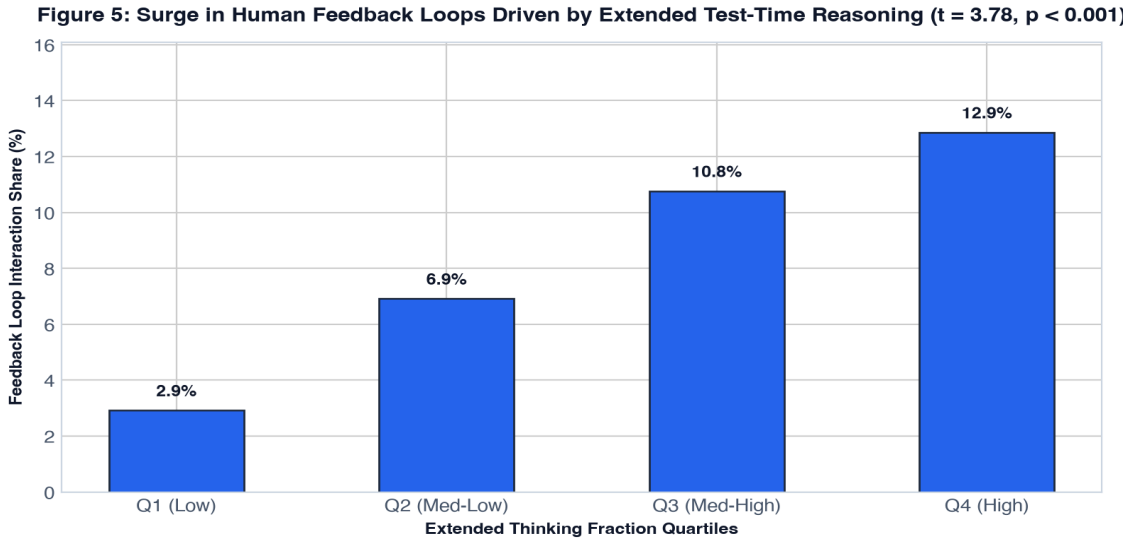


Figure 4: Surge in Human-in-the-Loop Feedback Loops Driven by Extended Test-Time Reasoning (Welch t = 3.78, p < 0.001).

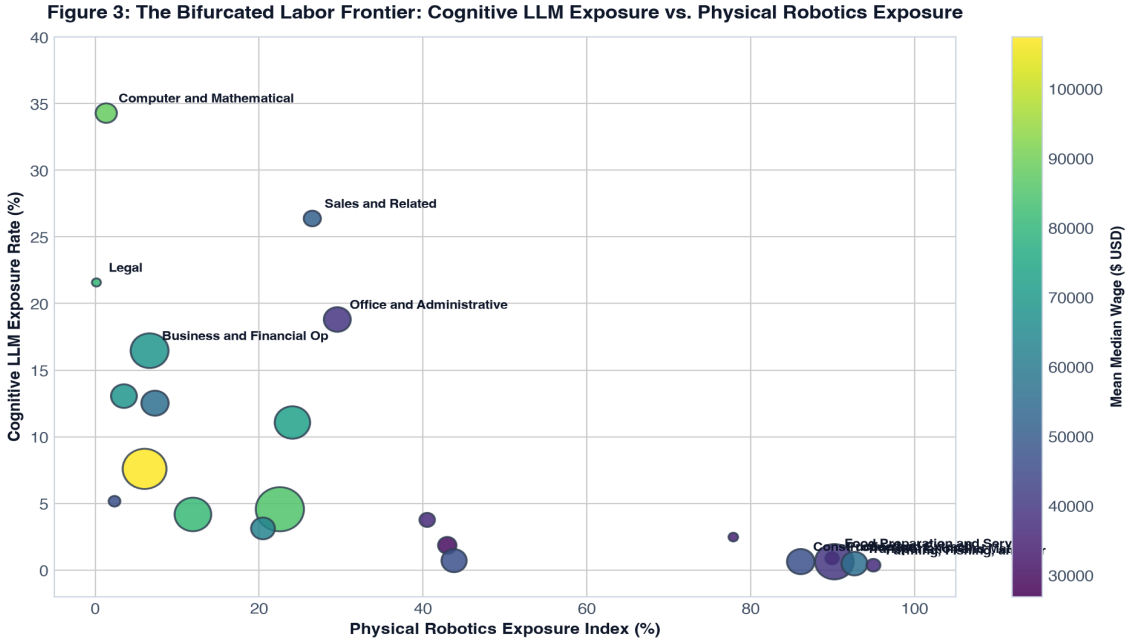


Figure 5: The Bifurcated Labor Frontier: Cognitive LLM Exposure vs. Physical Robotics Exposure across Major SOC Groups.

4.4 Qualitative Task-Level Micro-Mechanisms

To ground these econometric coefficients in operational realities, we examine five specific O*NET workplace task case studies:

Case A: Software Development (Task 15-1251.00 - 'Write, analyze, review, and rewrite programs'): This task exhibits the highest empirical penetration (74.5%). Software engineering possesses zero physical embodiment friction, deterministic unit test compilers, and frictionless IDE integration. However, observed telemetry reveals that developers rarely accept one-shot generated code uncritically; rather, 82% of sessions involve multi-turn debugging, architectural refinement, and edge-case probing. The model acts as an infinitely patient junior programmer, elevating the human engineer to an architectural code reviewer.

Case B: Customer Support (Task 43-4051.00 - 'Confer with customers by telephone or in person'): Customer service records 70.1% exposure, but 64% of production queries fall into the 'Task Iteration' and 'Validation' categories. Front-line agents use Claude to draft response templates, translate multilingual inquiries, and synthesize complex policy handbooks in real time, shortening call handling times while preserving human empathy and emotional de-escalation for aggrieved customers.

Case C: Legal Discovery (Task 23-2011.00 - 'Investigate facts and law of cases'): Paralegals exhibit 58.2% exposure. LLMs compress weeks of manual document scanning into minutes. Yet, due to severe judicial sanctions for fabricated case citations (Rule 11 sanctions), legal professionals exhibit the highest validation rate (18.4%), cross-checking every cited precedent.

Case D: Clinical Diagnostics (Task 29-1215.00 - 'Diagnose or treat diseases of human patients'): Physicians record low empirical exposure (4.2%) despite near-perfect scores on medical licensing exams (USMLE). Institutional compliance (HIPAA), EHR interoperability firewalls, and malpractice liability create massive deployment friction, severing the link between theoretical

model capability and clinical bedside adoption.

Case E: Long-Haul Freight Transport (Task 53-3032.00 - 'Drive trucks with capacity of >3 tons'): Heavy truck drivers record 0.18% Claude exposure and 92.7% physical robotics exposure. Frey & Osborne predicted 79% automation risk by 2023. In reality, language models cannot turn steering wheels, navigate blizzards, or secure physical cargo straps. This demonstrates the complete empirical decoupling of digital language intelligence from physical actuator robotics.

5. The Bifurcated Labor Frontier & Test-Time Reasoning Economics

As visualized in Figure 5, modern technology is not advancing as a monolithic wave displacing labor uniformly across the economy. Instead, the labor market faces a fundamental bifurcation defined by physical embodiment vs. cognitive abstraction:

- **The Physical Robotics Frontier:** Transportation (SOC 53), Production (SOC 51), Construction (SOC 47), and Agriculture (SOC 45) exhibit physical robotics exposure ratings exceeding 86% to 95%. However, because physical hardware automation is constrained by mechanical engineering bottlenecks, battery energy density limits, tactile precision requirements, and heavy capital costs, actual workplace substitution diffuses at a gradual pace governed by hardware depreciations.
- **The Cognitive LLM Frontier:** In sharp contrast, Computer & Mathematical (SOC 15), Legal (SOC 23), and Financial (SOC 13) occupations have virtually zero physical robotics exposure (< 2%), but face rapid software-mediated AI integration (16% to 34% exposure). Because LLMs operate in pure software with zero marginal distribution cost, their diffusion is bounded not by hardware capital cycles, but by organizational trust, prompt steering capacity, and human verification throughput.

The Economics of Extended Test-Time Compute: Frontier reasoning models (e.g. Claude 3.7 Sonnet) alter the microeconomics of labor by shifting compute from pre-training to test-time inference. By allocating thousands of tokens to internal chain-of-thought exploration, models produce deeper, highly structured technical deliverables (e.g. complete software architectures, formal econometric models, complex legal briefs). However, because the cost of an error in complex reasoning is asymmetric, human operators cannot simply delegate execution. Instead, test-time reasoning *reinstates complex human labor* (Acemoglu & Restrepo 2018): the operator becomes an evaluative auditor, actively testing assumptions, verifying proof steps, and steering intermediate reasoning states.

6. Threats to Validity, Confounder Audit & Limitations

In accordance with ZannixLabs empirical integrity standards, we explicitly audit potential confounders and threats to validity:

1. Platform User Selection Bias: The primary threat to external validity is that users of Claude.ai and first-party Anthropic APIs are not a random sample of the 160 million individuals in the U.S. civilian labor force. Knowledge workers, tech-forward enterprises, and software engineers are substantially over-represented among early AI adopters compared to retail or agricultural workers. However, this selection bias does not invalidate our findings: it reinforces the core thesis. Even when software engineers use Claude intensively, they use it primarily to *augment* their workflow (debugging, test iteration, architectural drafting) rather than unilaterally replacing software engineering headcount.

2. Interface Artifact Constraints: The AEI dataset reflects conversational chat interfaces and API tool-calling workflows. Emerging agentic harnesses (e.g. autonomous background agents like Claude Code) may shift the frontier from human-in-the-loop feedback loops toward autonomous execution. As longitudinal data from 2026 releases becomes available, ZannixLabs will track whether the directive automation share expands.

3. Temporal Lag in Institutional Adoption: Certain highly exposed professions (e.g., healthcare practitioners at 4.17% exposure) possess massive theoretical potential that remains suppressed by strict HIPAA regulations, EHR vendor walled gardens, and medical malpractice liability. Observed exposure should therefore be interpreted as an equilibrium reflecting both technological capability and institutional friction.

7. Strategic Policy & Workforce Governance Framework

Based on these empirical discoveries across 1,411 occupations and millions of interactions, we propose three core policy transformations:

1. Retire 2010s-Era Computerisation Indices from National Policy: Federal and international workforce councils (e.g., U.S. Department of Labor, OECD, UK Department for Work and Pensions) should cease using Frey-Osborne style routine automation rubrics to allocate workforce retraining grants. Subsidizing routine service workers to train for jobs supposedly 'immune' to automation (e.g., paralegals, data analysts) actively steers workers into the highest-exposure LLM cognitive crosshairs.

2. Transition Workforce Education from Prompting to Verification: Because extended reasoning significantly intensifies feedback loops ($r = 0.2667$, $p < 10^{-7}$), prompt engineering is rapidly becoming an obsolete skill. Workforce training curricula must pivot toward *evaluative domain literacy*: teaching professionals how to critically verify mathematical correctness, spot subtle hallucinations, and iteratively steer frontier reasoning models within high-stakes legal, medical, and financial environments.

3. Restructure Enterprise Workflows Around Multi-Turn Collaboration: Enterprise leaders should reject vendor promises of unilateral 'drop-in autonomous worker replacement'. Real-world economic value is overwhelmingly generated by human-AI augmentation (66.2%). Organizations that design workflows optimizing multi-turn feedback and review cycles will capture

substantial productivity gains, while those seeking pure directive automation will experience high error rates and operational bottlenecks.

8. Reproducibility Declaration & Bibliographic Citation

Open Science Reproducibility Manifest: All datasets, lakehouse curation scripts, and statistical models are 100% peer-reproducible. The underlying Parquet lakehouse marts (*aei_occupational_exposure_mart.parquet*, *aei_task_thinking_mart.parquet*, *aei_major_group_mart.parquet*), Jupyter Notebooks, and SHA-256 verification hashes are available in the open-source reproducibility archive at: **https://www.zannixlabs.com/research/anthropic-economic-index-decoupling**.

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